

Miguel Baidal Marco
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Currently based in
Prague (Czech Republic)

Social Media
LinkedIn Instagram

Miguel Baidal Marco

PhD Candidate in Computer Science | Biomedical Engineer | MSc Health Data Science | Pro Athlete

About Me I currently combine my PhD in Computer Science at the Czech Technical University in Prague with professional long-distance athletics (High-Level Athlete, Spanish 10K Champion, European U23 bronze, PB 28:17 in the 10,000 m).

My research focuses on artificial intelligence for health and sport: personalized models from wearable data, agentic AI, and the responsible evaluation of LLMs. I have experience in predictive modeling, machine learning, wearable data pipelines, and LLM evaluation in mental health (ACL 2025, JMIR Mental Health 2026).

Education

2026 – Present, Czech Technical University in Prague

Full-time PhD in Computer Science (Faculty of Electrical Engineering).
Thesis: "Agentic AI for Personalised Physiological Monitoring in Longitudinal Clinical Research"
Supervisor: doc. Ing. Petr Kadera, Ph.D.; supervisor specialist: Ing. Erik Derner, Ph.D.

2025 – 2026, Eastern Kentucky University, USA

Sport Management studies. NCAA Division I student-athlete.

2023 – 2025, Universitat Rovira i Virgili

Master's Degree in Health Data Science (taught entirely in English, 120 ECTS). GPA: 8.64/10.
Master's Thesis (Excellent, 9.0): "Evaluating Classical and Novel Machine Learning Approaches for Early Alzheimer's Detection Using fMRI Data"
Supervised by Dr. Roser Sala Llonch (Institute of Neurosciences, University of Barcelona)

2019 – 2023, University of Alicante

Bachelor's Degree in Biomedical Engineering. GPA: 8.21/10.

Work Experience

2026 – Present, Personal Project: GenesysLab

- Developing, together with two Italian collaborators, a platform for endurance coaches that brings together planning, athlete management, and automatic training-load analysis.
- Currently working on an individualized predictive model of each athlete's response to training.

July 2025 – October 2026, Health Data Scientist at Tolion Health Inc.

- Collected, integrated, and analyzed sports and health data from wearable devices (e.g., Garmin) used by the company's clients.
- Developed data pipelines and predictive models to evaluate the impact of users' actions on their health, with a special focus on potential risks related to dementia.
- Contributed to the frontend and backend development of the platform.

February 2025 – June 2025, Internship at ELLIS Alicante

- Authored my first scientific paper, "Guardians of Trust: Risks and Opportunities for LLMs in Mental Health," accepted at the 4th Workshop on NLP for Positive Impact (ACL 2025), with Nuria Oliver and Erik Derner [2].
- Proposed a taxonomy of risks and an evaluation framework for LLMs in mental health, and a research agenda for safer, clinically aligned AI systems.



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- Co-developed an evaluation framework for LLMs in crisis detection and intervention: dataset creation, crisis category definition, and model benchmarking. Published in JMIR Mental Health (2026), second author [1].

April 2022 – July 2022, Internship at Grupo Human Robotics, UA

- Early detection of Parkinson's disease through the design and analysis of physical-motor variables using a Unity-based VR game. Final grade: Outstanding (10).
- Skills: physical-motor variable design, application development (Unity, Matlab, Python), TCP/IP communication, automated data storage.

October 2021 – May 2022, Extracurricular Athletics Instructor

- Athletics coach for children aged 6 to 12 at Voramar Public School, Alicante.

Scientific Publications

[1] Adrián Arnaiz-Rodríguez, Miguel Baidal, Erik Derner, Jenn Layton Annable, Mark Ball, Mark Ince, Elvira Pérez Vallejos, Nuria Oliver. "Between Help and Harm: An Evaluation Study of Mental Health Crisis Handling by Large Language Models." *JMIR Mental Health 13, e88435 (2026)*. Link to article

[2] Miguel Baidal, Erik Derner, Nuria Oliver. "Guardians of Trust: Risks and Opportunities for LLMs in Mental Health." *Fourth Workshop on NLP for Positive Impact (NLP4PI), ACL 2025, pp. 11–22*. Link to article

Languages

Spanish

Native language

English B2 (certified)

Cambridge First Certificate. Current level above the certificate: one year living and studying in the United States and a PhD conducted entirely in English.

French A2

Valencian C1

Certifications

2022, IBM Data Science Professional Certificate (IBM / Coursera)

Ten courses: Python, SQL, data analysis and visualization, machine learning, and an applied capstone project. Verify

Awards and Distinctions

2024, Best University Student-Athlete Award of the Valencian Community

Awarded in recognition of outstanding academic performance and athletic achievements throughout the entire university period.

2023, URV Master's Scholarship for the Master's Degree in Health Data Science

Granted based on the academic record of the Bachelor's Degree and the curricular pathway so far (language level, work experience, awards and scholarships, etc.).

2023, Member of the FER Project

2021, 2022, 2023, Best Student-Athlete Award at the University of Alicante

Based on sports and academic results from the previous year.



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2023, Grade 10 in the Bachelor's Thesis in Biomedical Engineering
"Monitoring the degree of Parkinson's development using VR"

Major Athletic Achievements

- PB: 28:17 in 10,000 m – 13:40 in 5,000 m – 7:57 in 3,000 m
- 9th NCAA Division I Outdoor Championships, 10,000 m (2026), All-American honors
- 2nd Ibero-American Championships, 10K road (2024)
- 1st Spanish Senior Championships, 10K road (2024)
- 2nd World University Cross Country Championships (2024)
- 3rd U23 European Championships, 10,000 m (2023)
- 1st Spanish U23 Championships, 5,000 m (2022)
- 10 times international with the Spanish National Team:
  - 4th Mediterranean Games, 10,000 m, Taranto, Italy (2026)
  - 8th World University Games, 10,000 m, Bochum, Germany (2025)
  - 2nd Ibero-American Championships, 10K road, Cuiabá, Brazil (2024)
  - 38th World Athletics Cross Country Championships, Belgrade, Serbia (2024)
  - 2nd World University Cross Country Championships, Muscat, Oman (2024)
  - 5th U23 European Cross Country Championships, Brussels, Belgium (2023)
  - 3rd U23 European Championships, 10,000 m, Espoo, Finland (2023)
  - 48th U23 European Cross Country Championships, Turin, Italy (2022)
  - 56th U20 European Cross Country Championships, Lisbon, Portugal (2019)
  - 66th World U20 Cross Country Championships, Aarhus, Denmark (2019)

Other

- Spanish driving license. Category B